Prompt Engineering Masterclass (Solution Architect Track)

This is one of the most important days in your AI learning journey because Prompt Engineering is the foundation for:

- ChatGPT
- Microsoft Copilot
- GitHub Copilot
- Copilot Studio
- Azure OpenAI
- AI Agents
- RAG Solutions

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1. What is Prompt Engineering?

Prompt Engineering is the process of creating effective prompts that help AI models generate better responses.

Example

Poor Prompt

Explain Azure.

Output:

Generic answer

Better Prompt

Act as an Azure Solution Architect.

Explain Azure Service Bus.

Include:

1. Architecture
2. Benefits
3. Drawbacks
4. Cost
5. Alternatives

Provide markdown output.

Output:

Detailed Architect-Level Response

Why Prompt Engineering Matters

Same AI model:

Bad Prompt

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Poor Output

Good Prompt

↓

Excellent Output

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2. Prompt Formula

Use this formula:

Role

+

Task

+

Context

+

Output Format

Example

Role:

Azure Solution Architect

Task:

Explain Azure Event Grid

Context:

Audience has 10 years of .NET experience

Output:

Markdown with Architecture Diagram

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3. Zero-Shot Prompting

No example is provided.

Example

Translate to French:

Hello World

Use Cases

- Simple tasks
- Quick answers
- General questions

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4. One-Shot Prompting

One example is provided.

Example

Input:

Good Morning

Output:

Bonjour

Translate:

Input:

How Are You

Output:

Benefits

- Better consistency
- Better formatting

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5. Few-Shot Prompting

Multiple examples are provided.

Example

Input:

Good Morning

Output:

Bonjour

Input:

Thank You

Output:

Merci

Input:

How Are You

Output:

Benefits

- Higher accuracy
- Better structure
- Better consistency

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6. Role-Based Prompting

Tell AI who it should behave as.

Example Roles

Act as an Azure Architect.

Act as a Project Manager.

Act as a Technical Interviewer.

Act as a Security Consultant.

Example

Act as an Azure Solution Architect.

Explain Azure Service Bus.

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7. Chain Of Thought Prompting

Ask AI to solve problems step by step.

Example

Think step by step.

How should I design a secure Azure OpenAI architecture?

Benefit

Improves:

- Reasoning
- Analysis
- Accuracy

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8. Structured Output Prompting

Force AI output into a specific structure.

Example

```
Provide:

1. Summary
2. Risks
3. Recommendations
4. Action Items
```

Useful For

- Meeting Summaries
- Architecture Reviews
- Status Reports
- Project Plans

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9. Enterprise Prompt Patterns

Architecture Design Prompt

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Act as an Azure Architect.
Design an enterprise document processing system using:
- Azure OpenAI
- Azure AI Search
- Azure Storage

Include:
- Architecture
- Cost
```

- Security
- Data Flow

Code Review Prompt

Act as a Senior .NET Architect.

Review the code.

Identify:

- Security Issues
- Performance Issues
- Code Smells
- Best Practices

Meeting Summary Prompt

Summarize this meeting.

Provide:

- Key Discussions
- Decisions
- Risks
- Action Items

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10. Prompt Optimization Techniques

Technique 1

Be Specific

Instead of:

Explain Azure

Use:

Explain Azure Service Bus

for enterprise integration.

Technique 2

Use Role

Act as Azure Architect.

Technique 3

Add Context

Audience:

.NET Developers

Technique 4

Request Format

Provide output as Markdown.

Technique 5

Ask for Examples

Include real-world examples.

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11. Azure OpenAI Playground

The Playground helps you:

- Test prompts
- Improve prompts
- Compare responses
- Experiment with AI models

Workflow

Prompt

↓

AI Model

↓

Response

↓

Improve Prompt

↓

Better Response